

Reinforcement Learning Fundamentals

Learning value from sampled interaction without a known transition model

Learning objectives

By the end of this lesson, you should be able to:

- 01 — Distinguish planning from reinforcement learning.
- 02 — Compare model-based and model-free learning.
- 03 — Define episode, bootstrapping, behavior policy, and target policy.
- 04 — Derive and compute a TD(0) update.
- 05 — Explain exploration versus exploitation.
- 06 — Distinguish on-policy from off-policy learning.

Learning uses samples where planning uses a model

An RL agent must improve estimates and behavior from transitions it actually experiences.

- The transition model may be unknown.
- Rewards can be delayed and noisy.
- Exploration changes which state-action pairs are observed.

Experience

Data collection depends on current behavior, so the agent influences its own evidence.

The TD error compares a sample with a bootstrap target

$$\text{delta}_t = R_{(t+1)} + \text{gamma } V(S_{(t+1)}) - V(S_t)$$

- $R_{(t+1)}$ is sampled immediate reward.
- $V(S_{(t+1)})$ estimates continuation.
- $V(S_t)$ is the current prediction.
- delta is a signed prediction error.

ENGINEERING IMPLICATION

For a terminal successor, the continuation term is zero.

Model-based and model-free RL learn different objects

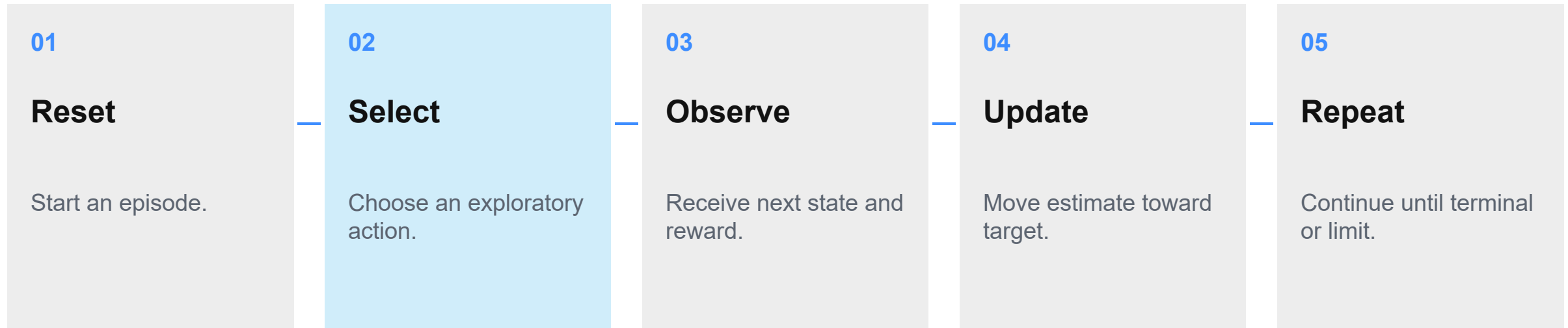
Model-based RL

- Learns or knows P and R .
- Plans inside the learned model.
- Can reuse simulated experience.
- Model bias can mislead.

Model-free RL

- Learns V , Q , or policy directly.
- No explicit transition table required.
- Often simpler control loop.
- May need more interaction.

Interaction creates an online learning loop



Behavior determines data coverage, so exploration is part of the learning algorithm.

TD(0) updates one transition at a time

```
for each episode:
    state = env.reset()
    while not terminal:
        action = behavior_policy(state)
        next_state, reward, terminal = env.step(action)
        target = reward
        if not terminal:
            target += gamma * V[next_state]
        error = target - V[state]
        V[state] += alpha * error
        state = next_state
```

Update controls

- alpha sets step size.
- gamma discounts future.
- Terminal stops bootstrap.
- Samples are correlated.

Worked TD update

SCENARIO

$V(A)=4$, transition A to B gives reward 2, $V(B)=5$, $\gamma=0.9$, and $\alpha=0.1$.

- $\text{Target}=2+0.9*5=6.5$.
- $\text{TD error}=6.5-4=2.5$.
- $\text{Increment}=0.1*2.5=0.25$.
- Set $V(A)=4.25$.

The estimate moves partway toward the one-step bootstrap target.

Planning and learning use different backup evidence

Bellman model backup

Full expectation

- Uses every successor probability.
- No sampled transition required.
- Deterministic given V .
- Requires known P and R .

Monte Carlo

Complete sampled return

- Waits until episode end.
- No bootstrap.
- Unbiased return sample.
- Can have high variance.

TD(0)

One-step sample

- Updates before episode end.
- Bootstraps from V .
- Lower variance, some bias.
- No model required.

Checkpoint: compute TD error

CHECKPOINT

$V(s)=3$, reward=-1, $V(s')=4$, gamma=0.5. What is the TD error and the new value for $\alpha=0.2$?

01

Compute the target.

02

Subtract the old prediction.

03

Apply the learning-rate step.

Exploration and exploitation serve different evidence needs

Exploitation

- Choose the current best action.
- Improves immediate performance.
- Concentrates on known behavior.
- Can lock into a poor estimate.

Exploration

- Try uncertain alternatives.
- Improves coverage.
- Can sacrifice immediate reward.
- Needs scheduling and limits.

Incremental updates average evidence over time

$$V(S_t) \leftarrow V(S_t) + \alpha_t [\text{target}_t - V(S_t)]$$

- α_t controls adaptation.
- Constant α tracks nonstationarity.
- Decaying α can support convergence.
- The bracketed term is prediction error.

ENGINEERING IMPLICATION

Large α learns quickly but can make estimates noisy or unstable.

Sparse reward exposes inadequate exploration

SCENARIO

A long corridor gives zero reward until a distant goal gives +10; an early side action gives +1.

- Greedy behavior discovers the side reward first.
- It repeatedly exploits +1.
- The distant goal remains unobserved.
- Exploration occasionally samples the corridor and reveals +10.

Observed performance depends on data coverage, not only the update equation.

On-policy and off-policy name the learning target

On-policy

Learn behavior being used

- Target includes exploratory policy.
- Evaluation matches data policy.
- SARSA is a control example.
- Safer behavior may be learned.

Off-policy

Learn a different target

- Behavior gathers data.
- Target may be greedy.
- Q-learning is an example.
- Coverage conditions matter.

Engineering

Name both policies

- Log epsilon schedule.
- Seed action sampling.
- Separate evaluation runs.
- Do not learn during testing.

Checkpoint: identify the policy relation

CHECKPOINT

An epsilon-greedy behavior policy generates data while updates target a purely greedy action. Is learning on-policy or off-policy?

01

Name the behavior policy.

02

Name the target policy.

03

Compare whether they are identical.

RL experiments need separation and repeatability

Separate training from evaluation, control randomness, and report variation across seeds.

- Evaluate with learning disabled and a fixed policy.
- Cap episode length to prevent hangs.
- Log return, length, exploration, and update magnitude.

Protocol

A rising training return is not an unbiased estimate of final policy quality.

Key takeaways

- 01 — RL learns from sampled interaction when the model is unavailable.
- 02 — TD combines one sampled reward with a bootstrapped continuation estimate.
- 03 — Exploration determines data coverage.
- 04 — On-policy and off-policy distinguish behavior from target.

Next: Q-learning applies off-policy TD updates directly to action values for control.

Check your understanding

Q1

Compute a TD(0) update.

Q2

Contrast model-based and model-free RL.

Q3

Explain bootstrapping.

Q4

Design an exploration schedule.

Q5

Distinguish behavior and target policies.